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Your Mitochondria and the Two Faces of Serotonin

Installment five in our series on understanding the truth about SSRIs.



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JUL 12, 2025



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In [our series](#) on how to understand the truth about serotonin, SSRIs, SSRI side effects, and the sometimes devastating effects of SSRI withdrawal, we have so far established that serotonin is a primarily non-brain non-neuronal signaling chemical that is fundamentally involved in the hypoxia response, and is also the substance from which we make melatonin, our mitochondria's guardian angel.

In the [last installment](#), we saw that serotonin is fundamentally involved in the whole-body response to hypoxia, beginning by acting as the traffic cop of our lungs, routing the blood flow away from the least oxygenated segments and toward the most oxygenated segments. This role is constant, because the oxygenation of our lungs is never completely homogeneous, so we need this to make sure our blood is always being efficiently oxygenated.

We are *always* dealing with *some* degree of hypoxic emergency *somewhere*, and our brain is the most disproportionate consumer of oxygen out of any tissue in our body. In fact, it makes up only two percent of our bodyweight but consumes twenty percent of our oxygen. So the fact that serotonin is mostly found outside the brain and is mostly acting on cells that aren't neurons does not in any way imply that serotonin is not important to brain function. The critical dependency of the brain on oxygen means that even this one role of serotonin — guiding the whole-body hypoxia response — will be critical to brain function.

We will now see that there are two faces of serotonin that each shine upon the mitochondria.

When serotonin is ***outside* the cell**, it stimulates the **production of new mitochondria** in a process known as **mitochondrial biogenesis**.

This is because serotonin is generally *activating* cells, which increases the demand for energy. For example, a single impulse through a neuron massively [increases the metabolic rate](#) of that neuron because the entire impulse is mediated by repeated fluxes of ions whose concentrations across membranes are maintained by ATP-dependent pumping. Serotonin causes smooth muscle cells in blood vessels to contract — in fact, serotonin was *named* after this effect — and muscle contraction requires energy because ATP is used up during muscle contraction. Mitochondria are what provide this ATP.

When serotonin is ***inside* the cell**, it acts directly on the mitochondria to **preserve mitochondrial respiration in the relative absence of oxygen**.

This is likely occurring because serotonin is the precursor to melatonin and melatonin possesses some unique direct ability to support mitochondrial respiration in the absence of oxygen. We will look at two possible explanations here:

- Melatonin is oxidized to a byproduct that directly substitutes for oxygen in the mitochondrial respiratory chain.
- Humans always produce ultraviolet and visible light. This may increase in hypoxia and melatonin may help translate shorter-wavelength light to longer wavelength light that can be absorbed by complex IV, converting it from an oxygen-using complex to a nitrogen-using complex, where the nitrogen is derived from the amino acid arginine.

Regardless of which explanation or mix of explanations is correct, serotonin and melatonin are both needed in concert to act directly on mitochondrial melatonin and serotonin receptors to signal the *capacity* to preserve mitochondrial respiration, and this enables the continued import of proteins into the mitochondria, prevents complex IV from being switched off by hypoxic regulation, and enables a flow of potassium ions into the mitochondria that preserves a critical currency of energy flow within the mitochondrial respiratory chain — the pumping of hydrogen ions. If any one of these ingredients is missing, mitochondria will “give up” on preserving respiration and the hypoxia response will then switch the cell toward using the much less efficient anaerobic glycolysis.

In areas of the body where serotonin is relatively abundant — such as in the gut or in many parts of the brain — a sudden increase in the metabolic demand could increase the hypoxia response and thereby increase serotonin transporters to bring serotonin into the cell for this purpose.

Many cells only experience extracellular serotonin, however, during the whole-body hypoxia response. Therefore, these cells are wired to interpret extracellular serotonin primarily as an early signal to ramp up the hypoxia response. In these cells, it is especially critical that the serotonin makes its way inside the cell, because this will be needed to balance the extracellular activation of the hypoxia response with the intracellular preservation of mitochondrial respiration. Otherwise extracellular serotonin will directly sacrifice the efficiency of cellular energy production by shifting the cell toward anaerobic glycolysis.

Extracellular Serotonin Activates Mitochondrial Biogenesis

Serotonin activates mitochondrial biogenesis [in rodent neurons](#) by binding to the 5-HT2A receptor through a cell signaling pathway involving phospholipase C, mitogen-activated protein kinase (MAPK), and sirtuin 1 (SIRT1).

In [kidney cells from rabbits](#), pharmacological activation of the 5-HT2B receptor stimulates mitochondrial biogenesis through a cell signaling pathway involving PGC-1alpha.

When mice have their genes altered to [overexpress the 5-HT2B receptor](#) specifically in the heart, they develop cardiac hypertrophy with excessive mitochondrial biogenesis,

although the mitochondria have defective transport of ATP and ADP across their membranes.

The promotion of mitochondrial biogenesis by 5-HT_{2B} receptors probably involves their stimulation of [nitric oxide production](#), a property they share with [5-HT_{1B} receptors](#) as well. Nitric oxide [promotes mitochondrial biogenesis](#) by activating guanylate cyclase, which converts guanosine monophosphate (GMP) to cyclic GMP (cGMP). cGMP activates protein kinase G, which activates PGC1-alpha.

[Activation of 5-HT_{2B}](#) in pancreatic beta-cells hurts glucose-induced ATP production and insulin secretion, which might be driven by PGC1alpha antagonizing beta-cell differentiation. The wiring of this signaling probably reflects the fact that increased mitochondrial content favors more fat burning and less glycolysis, and that glycolytic ATP production is what links glucose exposure to pancreatic insulin release.

Data conflicts on whether 5-HT_{2A} receptors [increase](#) or [decrease](#) nitric oxide, so its stimulation of mitochondrial biogenesis is more likely a result of classical activation of its G protein signaling cascade.

Most of these experiments tell us more about what specific serotonin receptors do than what the net effect of serotonin is, but they make it clear that activation of 5-HT_{2A} and 5-HT_{2B} receptors stimulates mitochondrial biogenesis, and that this can be one major effect of extracellular serotonin.

[5-HT_{2A} receptors](#) are found most abundantly in the brain and the sperm-producing cells of the testes. In the brain, they are most abundant in the excitatory neurons of the cerebral cortex, responsible for higher-level cognition; the cerebellum, involved in fine-tuning and course-correcting patterns during learning (not just balance and movement, but [cognitive patterns](#) as well); and the caudate, which is central to our use of dopamine to subconsciously calculate the value of investing energy in movement and changes in attention, emotion, and cognition. These receptors are the same as those thought to mediate the bulk of the effects of LSD, mescaline, and psilocybin.

Despite their greater abundance in part of the brain and in the male testes, they are found in every tissue measured. They are found in the cells that produce saliva, the glandular and epithelial cells of the breasts, gas-exchanging cells of the lungs, testosterone-producing cells of the testes, myelin-synthesizing and neuron-assisting cells of the brain, both skeletal muscle and smooth muscle cells, connective tissues, and skin cells.

[5-HT_{2B} receptors](#) are much more broadly expressed. Very few of them are found in the brain. Instead, they are abundant in tissues of the endocrine, reproductive, respiratory, digestive, and urinary systems, and in muscle, skin, bone marrow, lymph, pancreas, liver, and gallbladder.

Neurons are minor expressers of 5-HT_{2B} receptors. They are most abundant in connective tissue cells of the endometrium, the lining of the uterus. They have considerable expression in hepatocytes, the main type of liver cell; microglial cells, which are part of the brain's immune system; macrophages, immune cells involved in surveillance of tissues and removal of things that do not belong in the body;

fibroblasts, important cells in connective tissues; and endothelial cells, which make up the inner lining of blood vessels. They are also found in neurons, most glandular cells, keratin-producing skin cells, sperm precursors, adipose cells, immune cells, blood cells, cardiomyocytes (the main cells of heart muscle), and placental cells.

Extracellular Serotonin Can Stimulate the Cellular Hypoxia Response

The response to *sustained* hypoxia over the course of hours, or over the course of much longer periods of time, is mediated primarily by transcription factors called hypoxia-inducible factors (HIFs) that are present by default in the cytosol, the general area of the cell outside of specific organelles such as mitochondria.

Signals can travel from both the plasma membrane, which lines the outside of the cell, or from the mitochondria to activate the transcription factors, at which point they travel into the nucleus to regulate gene transcription. They also are involved in activating other proteins within the cytosol.

The action of serotonin on this system has been [attributed primarily to](#) activation of 5-HT_{1B} and 2B receptors on the cell surface, which can increase various sources of superoxide, hydrogen peroxide, and nitric oxide, all of which activate HIFs. This can also increase the quantity of HIF production by increasing transcription factors such as nuclear factor kappa B (NF- κ B), which plays broad roles in responding to stress, stimulating inflammation, and promoting cell survival. Thus, extracellular serotonin action on these receptors nearly immediately activates HIFs, and, if sustained, leads to the production of an even larger number of HIFs that can be activated.

Serotonin mediating this response makes perfect sense, since serotonin flux increases during hypoxia. However, if serotonin remains excessively elevated in the extracellular space, this would favor excessive activation of HIFs *in place of* the intracellular serotonin/melatonin axis activating the preservation of mitochondrial respiration. The net signaling of keeping serotonin in the extracellular space is to promote “giving up” on mitochondrial respiration as if it is “too late” and respiration must shut down in favor of anaerobic glycolysis.

These receptors are not by any means limited to the homeostatic oxygen-sensing tissues. The 5-HT_{1B} receptor [is found](#) mainly in the brain, gastrointestinal tract, and female reproductive tissues, with very light expression in most other tissues. In the brain, it is very broadly distributed, unlike the 5-HT_{2A} receptor.

The 5-HT_{2B} receptor is distributed as just described in the previous section.

We might speculate here that areas with higher expression of the 2A receptor than the 1B or 2B receptors — the cerebral cortex, cerebellum, caudate, and sperm-producing Sertoli cells of the testes — might be especially likely to increase mitochondrial biogenesis in response to extracellular serotonin rather than activate the hypoxia response.

The negative effects of 2B overactivation in cardiomyocytes and pancreatic beta-cells, moreover, might be confounded by the fact that the 2B receptors activate both mitochondrial biogenesis *and* the hypoxia response. Further, experimental

overactivation or overexpression of these receptors does nothing to leverage the ability of *intracellular* serotonin to balance this with the preservation of mitochondrial respiration during hypoxia.

In the gut, we might have a very well-blended effect of serotonin where 2B and 1B activation promotes the hypoxia response, 5-HT3 and 5-HT4 receptors promote gastrointestinal motility, and abundant serotonin enters cells to preserve mitochondrial respiration.

However, outside the gut and brain, most tissues are likely to *primarily* experience increased serotonin flux during hypoxia.

In this case, this impetus toward mitochondrial biogenesis and activation of the hypoxia response will *have* to be well balanced by *intracellular* serotonin to allow mitochondrial respiration to be preserved instead of prematurely switching to anaerobic glycolysis in order to gain maximum metabolic benefit.

Serotonin and the Mitochondrial Hypoxia Response

A study published in *Nature* last year showed that when [neurons are treated](#) continuously with serotonin for six days, it increases their ATP content without increasing their oxygen consumption and also increases the amount of acid that leaves the neurons into the medium.

This increase in the “extracellular acidification rate” was taken as a sign of increased glycolysis, and while this is widely interpreted as such in many studies, it is a totally dubious assumption for numerous reasons:

- First, [glycolysis is net alkaline](#) when it runs to completion. It only increases extracellular acidity if a) lactate leaves the cell, which causes a proton to be pumped out, but most lactate is oxidized in the cell’s mitochondria or b) glycolysis does not run to completion, in which the early acidic phase exceeds the later alkalizing phase.
- Second, the citric acid cycle produces CO₂, which is acidic.
- Third, as we will cover below, serotonin activates an influx of potassium into the mitochondrial matrix which increases proton pumping. There is evidence that this is partly mediated by potassium ions encouraging weak acids to deprotonate, which means that it would net increase the total protons available, which necessarily means an increase in acidity. Further, if the potassium drops in the cytosol, it could encourage taking up potassium ions from the medium, which would cause protons to leave the cell into the medium to maintain electric neutrality.

Thus, without actually measuring glycolysis, the rate of glycolysis remains unknown.

Instead, the most salient finding is that serotonin increases ATP production without increasing oxygen consumption. While this may have been from adding glycolysis on top of respiration, it may have been from enabling additional mitochondrial respiration without a proportional increase in oxygen consumption. This could be explained by the impact of *intracellular* serotonin on the hypoxia response.

Research in [mouse heart muscle cells](#) shows that mitochondria possess 5-HT₃ and 5-HT₄ receptors; in normoxia these receptors have very little if any net effect on mitochondrial metabolism; after twelve hours of zero oxygen, however, there are much more dramatic declines in ATP and increases in lactate dehydrogenase without these receptors. This suggests that the primary role of serotonin activating these receptors is to preserve mitochondrial ATP production during hypoxia.

The drop in ATP that occurs in hypoxia is due to the inability to make as much ATP without oxygen. The rise in lactate dehydrogenase is to respond to this drop in ATP production capacity with increased ability to engage in the less efficient anaerobic glycolysis. Since the combined effect of 5-HT₃ and 5-HT₄ receptors on mitochondria is to keep ATP content normal without requiring a rise in lactate dehydrogenase, it appears that activation of these receptors is serving to conserve respiratory chain ATP production despite drops in oxygen levels.

Serotonin May Have to Pass A Gate to Reach Mitochondria That Only Opens During Hypoxia

As to how activation of mitochondrial serotonin receptors would mediate ATP preservation during hypoxia but have no impact in normoxia, one possibility is that monoamine oxidase (MAO) could act as an oxygen-sensitive “gate” to prevent serotonin from accessing these receptors except during hypoxia. MAO degrades serotonin intracellularly using FAD and oxygen, and is embedded in the *outer* mitochondrial membrane. Serotonin receptors are embedded in the *inner* mitochondrial membrane. Anything traveling into the mitochondrion must first encounter the outer membrane before it arrives at the inner membrane. An acute drop in cellular oxygen would impair MAO activity and thereby open that gate. Incoming serotonin would communicate systemic hypoxia, while serotonin passing the MAO gate would reflect the cell’s own hypoxia. The mitochondria would thus get a combined signal that allows it to correctly anticipate whether the cell’s own immediate experience of hypoxia is a harbinger of worse things to come due to the systemic hypoxia.

The experiment above showed that the mitochondrial membrane serotonin receptors mediated the effect, so this excludes serotonin from having an effect *inside* the mitochondrial matrix, which lays beyond the inner membrane where the receptors are found.

However, serotonin does enter mitochondria, presumably through the serotonin transporter, and is [converted to melatonin](#) through an acetylation reaction dependent on acetyl CoA and a subsequent methylation reaction. The melatonin appears to then leave the mitochondrial matrix once formed and bind to cytosol-facing melatonin receptors on the outer mitochondrial membrane, which *also* protects against hypoxia through a mechanism that *also* involves restraining the expected increase in lactate dehydrogenase.

Inhibiting the melatonin receptors on the outer membrane *or* the serotonin receptors on the inner membrane abolishes this preservation of respiration during hypoxia, so it appears that all the receptors need to be activated in concert for this effect.

The Regulatory Effect of Serotonin and Melatonin on the Mitochondria

Given the need for the serotonin and melatonin *receptors* to engage this defense against hypoxia, it appears the effect should be *regulatory* in nature.

The impact of melatonin binding to its mitochondrial outer membrane receptor is to [inhibit cAMP production](#) specifically in the intermembrane space. cAMP production in the intermembrane space [inhibits the import](#) of proteins from the cytosol into the mitochondria and generally favors glycolysis over oxidative phosphorylation. Thus, binding of melatonin to its receptor would be expected to take the brakes off this import system, allowing proteins to enter the mitochondria during hypoxia to facilitate improved resilience of the respiratory chain, thereby preventing an overly rapid shift to anaerobic glycolysis.

The 5-HT₃ receptor is a channel that allows the free flow of sodium and potassium ions when it is opened by the binding of serotonin. In the inner mitochondrial membrane, this would be expected to cause potassium ions to enter the mitochondrial matrix. When potassium ions enter the mitochondrial matrix, they [improve ATP production](#), probably because their positive charge helps push hydrogen ions through proton pumps more effectively, which creates the H⁺ gradient that is used to synthesize ATP. It also increases the abundance of protons by helping acids to lose their proton into solution. In other words, it makes these acids more acidic. Together, a greater abundance of protons and a greater density of positive charge increases the proton pumping that drives respiratory chain activity.

The 5-HT₄ receptor at the inner membrane would be expected to increase cAMP in the mitochondrial matrix. In hypoxia, this might help maintain ATP production despite declining activity of the citric acid cycle.

Under ordinary circumstances, complex IV activity is [tied to the citric acid cycle](#) as follows. Accumulation of ATP inhibits complex IV due to there being sufficient ATP supply, while CO₂ produced in the citric acid cycle activates it, signaling that more energy is incoming into the respiratory chain. This avoids overproduction of unneeded ATP while also ensuring that any incoming electrons can be fully metabolized by complex IV to avoid any “traffic jams” in the respiratory chain.

The mechanism of this regulation is that CO₂ from the citric acid cycle converts to bicarbonate, which activates soluble adenylate cyclases in the mitochondrial matrix, which convert ATP to cyclic AMP (cAMP). cAMP then activates protein kinase A (PKA), which then phosphorylates complex IV, making it resistant to inhibition by ATP.

In hypoxia, the declining activity of the respiratory chain would impair the oxidation of NADH to NAD⁺ at complex I and the speed of the succinate dehydrogenase reaction (which links the citric acid cycle to the respiratory chain via complex II), and this would slow the generation of carbon dioxide by the citric acid cycle, thus lowering the production of cAMP in the mitochondrial matrix and thereby making complex IV more sensitive to inhibition by ATP. Activation of the 5-HT₄ receptor would provide

an alternative means of producing cAMP in the mitochondrial matrix and keep complex IV in its more active state.

Overall, serotonin and melatonin lie at the crossroads of a decision to respond to hypoxia either by fortifying the capacity for mitochondrial respiration under these conditions in their presence, or by “giving up” on mitochondrial respiration and shifting toward anaerobic glycolysis in their absence.

The Mysterious Ability of Melatonin to Support Respiration Without Oxygen

The question then arises of *why* serotonin and melatonin would exert these regulatory roles. It makes little sense that the presence of serotonin or melatonin should create a bifurcation in the signaling pathway for handling hypoxia unless their presence is actually *causing* in some other way a greater *capacity* to rely on mitochondrial respiration to produce energy. Otherwise, they would simply be hurting the ability to respond properly to hypoxia by inhibiting a necessary rise in anaerobic glycolysis.

It may be that serotonin passing through the “MAO gate” is the primary determinant of how much serotonin enters the mitochondria for melatonin production, and that the availability of serotonin and melatonin to the mitochondria are thus essentially the same thing, providing there is sufficient acetyl CoA and methyl groups available to convert the serotonin to melatonin. While it is known that melatonin is an antioxidant, this is a rather dubious reason for it to stand out in singular capacity to prevent the need to respond to hypoxia with anaerobic glycolysis, simply because there are many other compounds that participate in a complex network of antioxidant defense.

One thing that does stand out about melatonin as a redox-reactive molecule is the [complexity of byproducts](#) it can form, which might lead to a multiplicity of specific roles, with one or more byproducts or some type of synergy between the byproducts being highly relevant during hypoxia.

One possibility is that after a single-electron oxidation of melatonin produces the [melatonyl cation radical](#), this radical could then oxidize cytochrome C, serving to displace the role of oxygen in complex IV. This would allow continued oxidation of NADH by complex I without the role of oxygen in complex IV, and would allow the continued pumping of protons by complexes I and III while losing the oxygen-dependent pumping of protons by complex IV. Although the activity of oxygen at complex IV is what usually allows the electrons to flow through the beginning of the chain, complex IV only directly pumps 20% of the protons within the chain. Thus, replacing its activity with that of melatonin would allow the rest of the chain to function and conserve 80% of ATP generation. By allowing the oxidation of NADH, it would keep the citric acid cycle going as well.

Another possibility would be that melatonin or one of its oxidation products is funneling photons into complex IV. Humans always emit light, and this is [thought to be](#) a by product of reactive oxygen species. During hypoxia, nitric oxide production from the amino acid arginine increases. Nitric oxide is a free radical that can lose electrons to generate nitrite in a number of different ways. Nitrite can [replace oxygen](#) at complex IV to have electrons added back to it, reforming nitric oxide. This is

dramatically enhanced by light in the yellow-orange range (590 nm). While this was shown to be superior to red light (627 or 660 nm), it was not compared to blue light. Melatonin [shifts light](#) redward, absorbing ultraviolet light and releasing it between 300 (UV) and 550 (green) nm, with large portions in the violet (400) to blue (450) nm range. Its [major oxidation product](#) is not very different. Humans, however, only emit a [small amount](#) of ultraviolet light and emit the most light in the green range, which is quite close to the light shown experimentally to increase the nitrite reduction capacity of complex IV. This casts doubt on whether the wavelength-shifting effect melatonin would be very helpful.

However, the effect of hypoxia on the emissions spectra of human cells has not been studied, UV is hard to detect because much of it is absorbed internally, whether blue light supports nitrite production rather than just yellow-orange light has not been studied, and the absorption/emission spectra of a variety of melatonin oxidation products has not been studied.

Thus, I would remain agnostic right now as to which of these are true and say only that we need desperately to study both potential explanations in more detail, but I do personally lead toward the first hypothesis that the melatonyl cation radical is oxidizing cytochrome C and preserving 80% of the energy production.

The Overall Effect of Serotonin In Hypoxia

The release of serotonin from the first oxygen-deficient breath would immediately raise the breathing rate and begin rewiring the blood supply to maximize tissue oxygenation. Hypoxic cells will increase their serotonin transporters to bring serotonin inside the cell, and lack of oxygen will open the monoamine oxidase “gate,” allowing serotonin to reach the mitochondria. Mitochondrial serotonin would first signal to the mitochondria not to give up importing fortifications of the respiratory chain or prematurely turning down the activity of complex IV, and then would enter the mitochondria and be converted to melatonin. Some of the melatonin would exit the mitochondria to further signal that surrender is not necessary, while the rest would serve as a *substitute* for oxygen to the extent such a substitute is necessary, allowing at least 80% of the ATP to be generated with each substituted round of electron transport. When the magnitude of cellular hypoxia exceeds this serotonin/melatonin flux, the cell will finally “give up” on fortifying the mitochondria and shift to anaerobic glycolysis until oxygen levels can be restored.

The Overall Effect of Serotonin On Mitochondrial Function

The effect of serotonin on the mitochondrial metabolism of any given tissue or cell reflects the degree to which it is present, the mix of extracellular 5-HT receptors expressed, the expression of the serotonin transporter, and the expression of mitochondrial 5-HT receptors.

In the cerebral cortex, cerebellum, caudate, and sperm-producing cells of the testes, we are likely to get a greater stimulus toward mitochondrial biogenesis without as much stimulation of the HIF-mediated hypoxia response.

This makes extracellular serotonin especially important to mitochondrial biogenesis where we need it for higher-order cognition; course-correcting balance, movement, and emotional and cognitive states during learning from experience; using dopamine to subconsciously calculate the value of investing energy in movement, and the shifting of attention or emotional and cognitive states; and, in males, sperm production.

Even in these cells, serotonin is likely to increase energy production and its entry into the cell will allow a greater increase without a proportional need for more oxygen.

In the gut, we are likely to get a well-mixed blend of mitochondrial biogenesis, all aspects of the hypoxia response, and gastrointestinal motility.

In most cells, we primarily use serotonin toward these purposes during whole-body hypoxic stress. This is relevant not only during hypoxic stress from improper breathing, anemia, or altitude, but also due to increased oxygen demand during intense cognition or exercise. Under these conditions, we get a blend of hypoxia-mediated responses that include activations of HIFs but also preservation of mitochondrial respiration due to the intracellular actions of serotonin and melatonin.

Whereas the baseline mitochondrial melatonin content is probably from uptake of melatonin secreted by the pineal gland at night, mitochondria massively increase their melatonin content during hypoxia by using serotonin taken in through the serotonin transporter.

This explains the observation I made in [Melatonin Is Your Mitochondria's Guardian Angel](#) that removal of the pineal gland in rats dramatically increases intracellular melatonin. Removing the pineal gland causes circadian stress wherein cells cannot use proper anticipation to match their respiratory capacity to the ebb and flow of oxygen consumption across the day, which winds up producing an absolutely massive [26-fold elevation](#) of HIFs. That is, an absolutely massive increase in the cellular hypoxia response. This increases the serotonin transporter, taking serotonin into the cell so that the mitochondria can convert it to melatonin.

Whenever serotonin acts on a cell, we *need* it to be able to enter the cell. If we stop it from doing this, we will get an imbalanced hypoxia response where HIFs are activated and mitochondrial respiration is not preserved. HIFs then switch the cell from mitochondrial respiration to the much less efficient anaerobic glycolysis.

Now, you may be thinking, so *this* is why SSRIs are playing with fire.

As we will see in the next installment, that is part of it. But, in fact, SSRIs also enter cells, where some of them are powerful activators of receptors that have almost nothing to do with serotonin and then have their own direct impacts on mitochondria. So actually, the fire is a lot harder to handle than it looks. We examine this in the next installment. You can read it here:

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SSRIs are Mitochondrial Drugs

CHRIS MASTERJOHN, PHD • JUL 13



Our series has so far focused on serotonin and melatonin but now turns to SSRIs themselves. These not only interact with the ability of serotonin and melatonin to support mitochondrial function in both good and bad ways, but are themselves drugs that act on mitochondria both by activating receptors that have little to do with serotonin inside cells and ...

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I would really like to know if people heal once off. And what is required. Thank you for this discussion on ssris. This all makes sense to me

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